SIMATS ENGINEERING

ASSESSMENT TOOL 1: Knowledge Representation Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE OUTCOME COVERED
COURSE CODE: MLA0104

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3) Assessment Tool Weightage: 40 %

Total Marks: 100 **Time: 60 Mins** **No. of Questions: 5**

Annexure A

Knowledge Representation Design Assessment

Code of Conduct:

*I , **M. Sanjay Krishna/192525421** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the student: M. sanjay

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Knowledge Engineering and Knowledge Base Design	15%	
3. Propositional Logic Representation	10%	
4. First-Order Logic Representation	15%	
5. Unification and Resolution	15%	
6. Forward and Backward Chaining	15%	
7. Inference and Reasoning Accuracy	10%	
8. System Architecture / Representation Diagram	5%	
9. Prototype Implementation and Results	5%	
Total	100%	

1. Intelligent Loan Approval System

A bank wants to develop an AI system to determine whether a customer is eligible for a loan. The knowledge base contains information about income, credit score, employment, existing loans, and repayment history.

For example:

- Arun has a stable job.
- Arun has a high credit score.
- Arun has sufficient income.
- Customers with stable employment and sufficient income are eligible for preliminary approval.
- Customers with high credit scores and good repayment history are considered low-risk.
- Low-risk customers with preliminary approval can be recommended for loan approval.

Task:

1. Identify the **entities, facts, predicates, and rules** involved.
2. Represent the knowledge using **First-Order Logic**.
3. Build a knowledge base with at least **10 facts and 8 rules**.
4. Demonstrate **unification and substitution**.
5. Apply **forward chaining** to determine whether Arun can be recommended for approval.
6. Apply **backward chaining** for the query Loan Approved (Arun). 7. Use **resolution** to prove or disprove one query.
8. Discuss how the knowledge-engineering process helps maintain the loan rules.

Intelligent Loan Approval System

1. Problem Analysis and Domain Understanding

Problem

A bank wants an AI-based system to decide whether a customer is eligible for a loan using income, employment, credit score, existing loans, and repayment history.

Objective

To determine whether Arun can be recommended and approved for a loan using knowledge representation and reasoning.

Entities

- Customer
- Income
- Employment
- Credit Score
- Existing Loan
- Repayment History
- Risk
- Loan Approval

2. Knowledge Engineering and Knowledge Base

Predicates

Predicate	Meaning
Stable Job(x)	x has stable employment
High Credit(x)	x has high credit score
Sufficient Income(x)	x has sufficient income
Good Repayment(x)	x has good repayment history
No Default(x)	x has no loan default
Preliminary Approval(x)	x gets preliminary approval
Low Risk(x)	x is low-risk
Good Profile(x)	x has a good financial profile
Eligible(x)	x is eligible
Loan Recommended(x)	x is recommended
Loan Approved(x)	x is approved

Facts:

1. Stable Job (Arun)
2. High Credit (Arun)
3. Sufficient Income (Arun)
4. Good Repayment (Arun)
5. No Default (Arun)
6. Manageable Debt (Arun)

7. Stable Job (Priya)

8. Sufficient Income (Priya)

9. High Credit (Rahul)

10. Good Repayment (Rahul)

11. Existing Loan (Meena)

12. High Risk (Meena)

Rules

R1. Preliminary Approval
Stable Job + Sufficient Income → Preliminary Approval

R2. Low Risk
High Credit + Good Repayment → Low Risk

R3. Good Profile
Sufficient Income + No Default → Good Profile

R4. Eligibility
Preliminary Approval + Good Profile → Eligible

R5. Existing Loan
Existing Loan + Manageable Debt → Eligible

R6. Recommendation
Low Risk + Preliminary Approval → Loan Recommended

R7. Final Approval
Loan Recommended + Eligible → Loan Approved

R8. Strong Profile
High Credit + Sufficient Income + Good Repayment → Eligible

3. Propositional Logic Representation

Let:

- P = Stable job
- Q = Sufficient income

- R = High credit
- S = Good repayment
- T = Preliminary approval
- U = Low risk
- V = Eligible
- W = Recommended
- X = Loan approved

4. First-Order Logic Representation

Facts:

Stable Job (Arun)

High Credit (Arun)

Sufficient Income (Arun)

Good Repayment (Arun)

No Default (Arun)

Rules:

∀x [Stable Job(x) ∧ Sufficient Income(x) → Preliminary Approval(x)]

∀x [High Credit(x) ∧ Good Repayment(x) → Low Risk(x)]

∀x [Sufficient Income(x) ∧ No Default(x) → Good Profile(x)]

∀x [Preliminary Approval(x) ∧ Good Profile(x) → Eligible(x)]

∀x [Low Risk(x) ∧ Preliminary Approval(x) → Loan Recommended(x)]

∀x [Loan Recommended(x) ∧ Eligible(x) → Loan Approved(x)]

Final Conclusion:

Loan Approved (Arun) = TRUE

5. Unification and Substitution

Consider the following two expressions:

Stable Job(x)

Stable Job (Arun)

Step 1: Identify the variable

The variable is **x**.

Step 2: Match x with Arun

Substitute:

$\theta = \{x / Arun\}$

Step 3: Apply the substitution

Stable Job(x) [x / Arun]

= Stable Job (Arun)

Therefore:

Stable Job(x) = Stable Job (Arun)

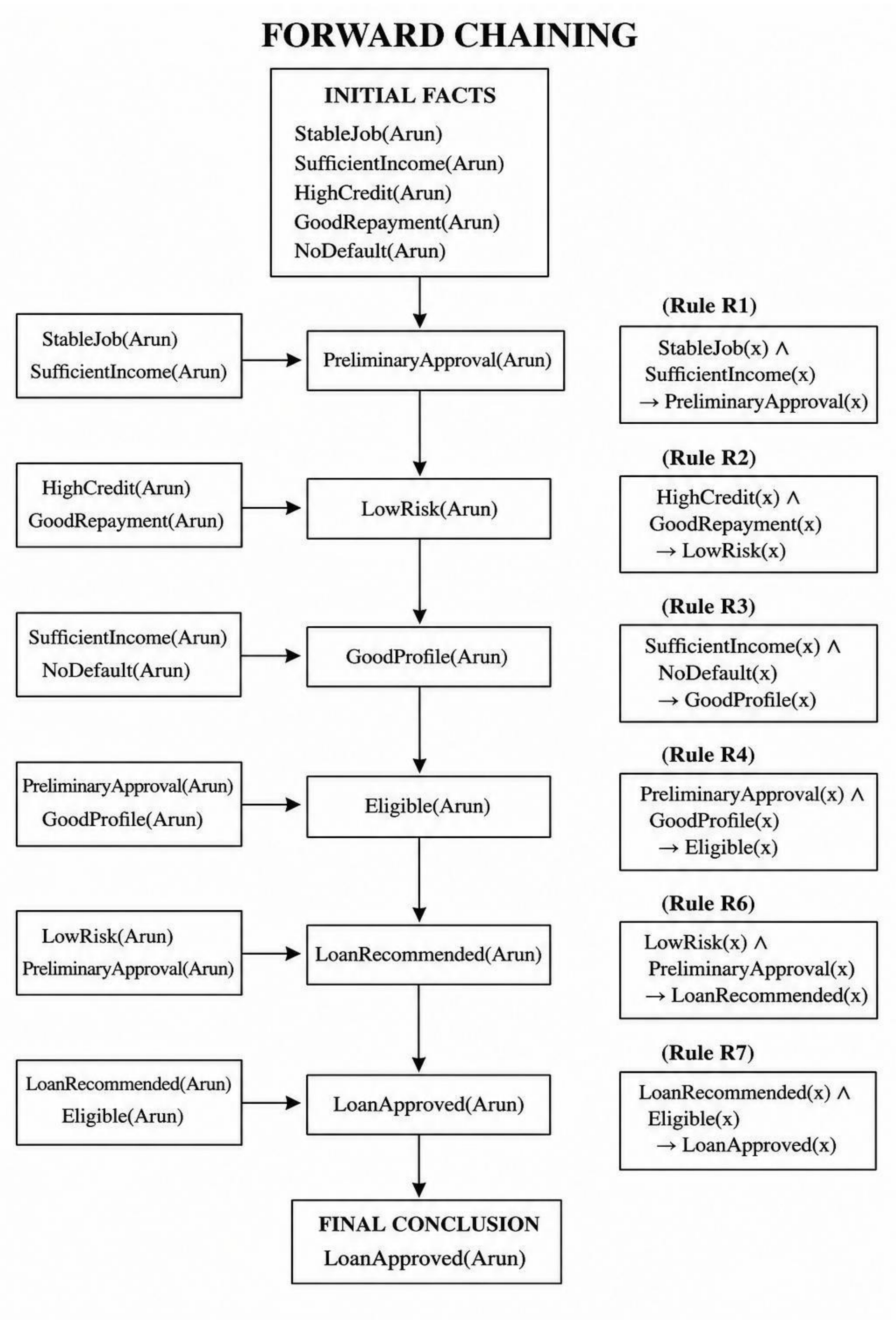
Most General Unifier (MGU):

$\theta = \{x / Arun\}$

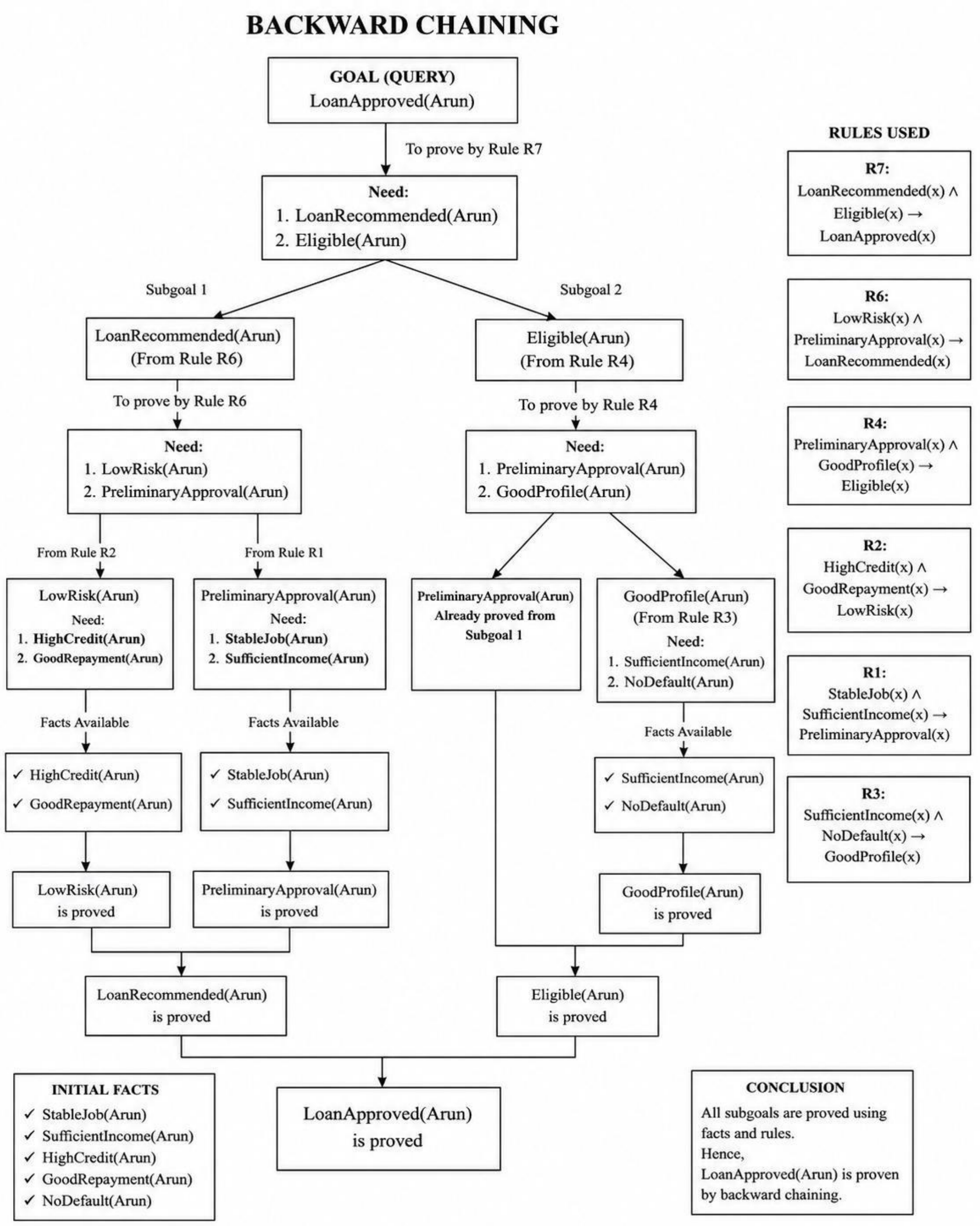
Result:

The variable **x** is replaced by **Arun**, making both expressions identical.

6. Forward Chaining



7. Backward Chaining



8. Resolution

Query:

Loan Approved (Arun)

1. Negate the query:

¬Loan Approved (Arun)

2. Convert Rule R7 to clause form:

¬Loan Recommended(x) ∨ ¬Eligible(x) ∨ Loan Approved(x)

3. Substitute x = Arun:

¬Loan Recommended (Arun) ∨ ¬Eligible (Arun) ∨ Loan Approved (Arun)

4. Known facts:

Loan Recommended (Arun)

Eligible (Arun)

5. Resolution gives:

Loan Approved (Arun)

This contradicts:

¬Loan Approved (Arun)

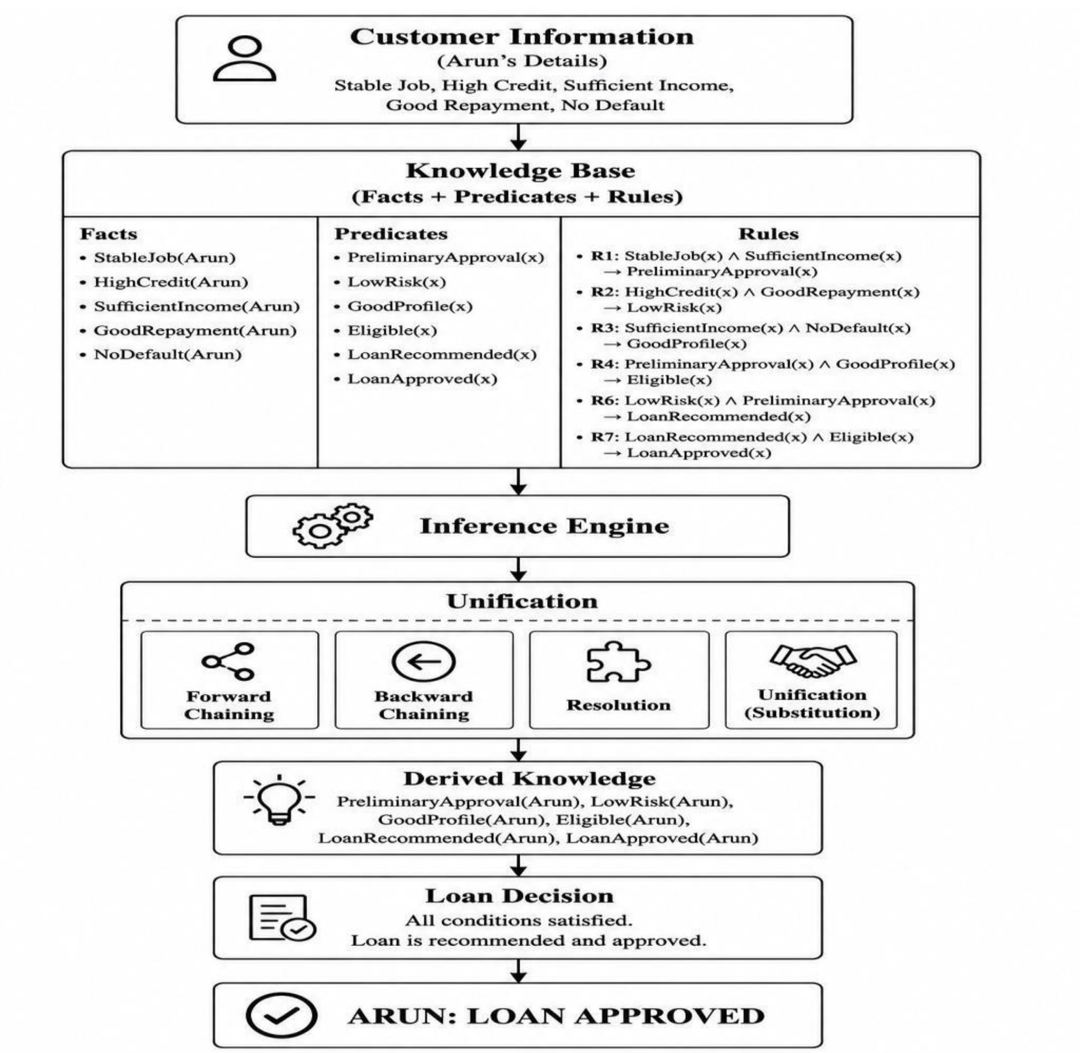
Final Result:

Loan Approved (Arun) is **PROVED**.

9. Inference and Reasoning Accuracy

Evidence	Inference
Stable job + sufficient income	Preliminary approval
High credit + good repayment	Low risk
Income + no default	Good profile
Preliminary approval + good profile	Eligible
Low risk + preliminary approval	Recommended
Recommended + eligible	**Loan approved**

10. System Architecture



11. Prototype Result

INTELLIGENT LOAN APPROVAL SYSTEM

Customer : Arun

Stable Employment : YES

High Credit Score : YES

Sufficient Income : YES

Good Repayment : YES

No Previous Default : YES

Preliminary Approval : YES

Risk Level : LOW

Eligibility : YES

Recommendation : YES

Final Approval : YES

RESULT: ARUN IS APPROVED FOR THE LOAN

12. Knowledge Engineering and Maintenance

Knowledge engineering helps the system by:

1. Collecting rules from banking experts.
2. Converting banking policies into predicates and rules.
3. Storing facts and rules in the knowledge base.
4. Applying inference methods to make decisions.
5. Testing the rules for correctness.
6. Updating rules when banking policies change.

13. Critical Comparison

Method	Approach	Use
Forward Chaining	Facts → Goal	Derives all possible conclusions
Backward Chaining	Goal → Facts	Tests a specific query
Unification	Matches expressions	Finds substitutions
Resolution	Proof by contradiction	Validates conclusions

Using multiple reasoning methods makes the system more accurate, explainable, and reliable.

14. Final Result

Inference Rules

1. Preliminary Approval

Stable Job + Sufficient Income → Preliminary Approval

2. Low Risk

High Credit + Good Repayment

→ Low Risk

3. Eligibility

Preliminary Approval + Good Profile

→ Eligible

4. Loan Recommendation

Low Risk + Preliminary Approval

→ Loan Recommended

5. Final Loan Approval

Loan Recommended + Eligible

→ Loan Approved

Final Flow

Stable Job + Sufficient Income

→ Preliminary Approval

→ Loan Recommended

→ Loan Approved

Final Conclusion

Arun is eligible, low-risk, recommended, and approved for the loan.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Analysis and Understanding		13–15% – Clearly understands and analyses the real-world problem, objectives, entities, facts, constraints, and expected inference.	10–12% – Understands the problem with minor omissions.	7–9% – Shows partial understanding with some important elements missing.	0–6% – Poor or incorrect understanding of the problem.
Knowledge Engineering & Knowledge Base Design		15% 13–15% – velops a complete, consistent, and well-structured knowledge base with appropriate facts, predicates, entities, and rules.	10–12% – Knowledge base is well designed with minor inconsistency es or omissions.	7–9% – Basic knowledge base is developed but several facts/rules are missing or unclear.	0–6% – Knowledge base is incomplete, inconsistent, or inappropriate.
Propositional & First-Order Logic Representation		20% 17–20% – rectly represents the domain using propositions, predicates, variables, constants, quantifiers, and logical connectives with excellent accuracy.	13–16% – Logic representation is mostly correct with minor errors.	9–12% – Provides basic representations but has several logical or syntactic errors.	0–8% – Logic representation is incorrect, incomplete, or missing.
Unification & Resolution		15% 13–15% – orrectly performs substitutions, identifies the MGU, and applies resolution systematically to derive valid conclusions.	10–12% – Applies unification and resolution correctly with minor errors.	7–9% – Demonstrates partial understanding but misses important steps or contains errors.	0–6% – Unable to correctly apply unification or resolution.

Forward Chaining & Backward Chaining		15% 13–15% – rrectly applies both ward and backward chaining with clear step-by-step reasoning and appropriate rule selection.	10–12% – Correctly applies both methods with minor errors or omissions.	7–9% – Demonstrates partial understanding of one or both chaining methods.	0–6% – Incorrectly applies or fails to demonstrate chaining techniques.
Inference & Reasoning Accuracy		10% 9–10% – roduces logically valid conclusions and clearly explains how facts and rules lead to the final inference.	7–8% – Conclusions are mostly correct with minor reasoning gaps.	5–6% – Some correct inferences but reasoning is incomplete or unclear.	0–4% – Conclusions are in correct, unsupported, or missing.
Knowledge Representation Diagram / System Architecture		% 5% – Provides a complete, accurate, ell-labelled diagram showing knowledge base, inference engine, reasoning process, and output.	4% – Diagram is mostly correct with minor omissions.	3% – Diagram is partially complete or lacks clarity.	0–2% – Diagram is missing or incorrect.
Originality, Critical Thinking & Presentation		5% 5% – Demonstrates excellent independent thinking, appropriate real-world application, comparison of reasoning approaches, and clear presentation.	4% – Demonstrates good reasoning and clear presentation.	3% – Shows limited critical thinking and basic presentation.	0–2% – Shows little originality, weak reasoning, or poor presentation.